

Introduction to mathematics and optimization

January 2025

The exam solutions must be based on the curriculum. Concepts and results in the solutions, which are not part of the curriculum, must be explained based on the curriculum.

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = (x + y)^2 - x - y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \{ (x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4, x + y \geq 1 \}.$$

We consider the optimization problem

$$\begin{aligned} \min f(x, y) \\ (x, y) \in C \end{aligned} \tag{1}$$

- (a) Prove that f is a convex function.
- (b) Explain why C is a bounded and convex subset of $\mathbb{R}^2$.
- (c) Show that C is compact and explain why this is important for the optimization problem (1).
- (d) Explain, based on the KKT conditions, that $(2, 0)$ is not an optimal solution to (1).
- (e) Show that $(1, 0)$ and $(0, 1)$ are optimal solutions to (1) based on the KKT conditions.
- (f) Is f a strictly convex function?

PROBLEM 2

Consider the system of linear equations

$$\begin{array}{rcrcrcrcl} 2x & + & y & = & 1 & & \\ x & + & 2y & = & t & & \end{array} \quad (2)$$

where $t \in \mathbb{R}$.

(a) The system of equations (2) can be written in matrix form $Av = b$, where

$$v = \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 1 \\ t \end{pmatrix}.$$

Find the matrix A .

(b) Explain why A is invertible and find A^{-1} .

(c) Explain why (2) has a unique solution for all $t \in \mathbb{R}$ and state the unique solution.

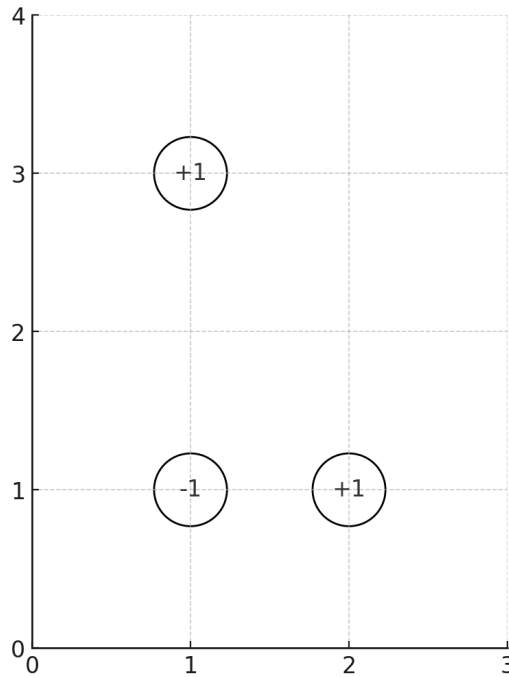
(d) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by

$$f(x, y, t) = (1 - 2x - y)^2 + (t - x - 2y)^2$$

It can be used without proof that f is convex. Find the critical points for f without solving the equations $\nabla f = 0$.

PROBLEM 3

- (a) The points $(1, 1), (1, 3), (2, 1) \in \mathbb{R}^2$ are provided with labels $-1, +1, +1$ respectively:



Show that if the line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ separates the points, then must

$$\begin{aligned} a + b + c &< 0 \\ a + 3b + c &> 0 \\ 2a + b + c &> 0 \end{aligned} \tag{3}$$

- (b) Show using Fourier-Motzkin elimination that a solution to the inequalities in (3) must satisfy $a > 0$ and $b > 0$, and that for given $a > 0$ and $b > 0$ there exists c such that the inequalities in (3) are satisfied.
- (c) Set up an optimization problem to find the best line that separates the points.
- (d) Show that $2x + y - 4 = 0$ is the best line that separates the points.
- (e) Is the solution to the optimization problem from sub-question (c) unique?